

Master's Mathematics

2026-27

Remark 1: This overview is intended to help guide your elective choices, but it does not carry official status. Although it has been prepared with care, the 2026-27 TER takes precedence over it.  
Remark 2: To students who started before 2026-27, transitional provisions apply. Discuss with the Study Advisor and/or see the 2026-27 TER, Article 11.2.  
Notation: 'p1-p6' denote the periods for VU-taught courses, while 'fall/spring' is used for Mastermath courses.

| COMPULSORY components                                                           |          | 54 or 60 EC |  |
|---------------------------------------------------------------------------------|----------|-------------|--|
| Course Name                                                                     | Code     | EC          |  |
| p1+2 Current Issues in Mathematics                                              | XM_0202  | 6           |  |
| Master Project Mathematics                                                      | X_400355 | 36          |  |
| Contrained choice in seminars: choose 2 or 3 throughout studies*                |          |             |  |
| p1+2 Data Assimilation**                                                        | XM_0203  | 6           |  |
| p1+2 Sequential Monte Carlo Methods                                             | XM_0182  | 6           |  |
| p4+5 Geometry of Statistical Models                                             | XM_0204  | 6           |  |
| p4+5 Covering Spaces in Topology and Algebra                                    | XM_0205  | 6           |  |
| *offer may vary from year to year                                               |          |             |  |
| **not in combination with XM_0183 Uncertainty Quantification offered in 2025-26 |          |             |  |

Mathematics and Data study path

| STRONG RECOMMENDATIONS for electives                 |            | ≥30 EC |  |
|------------------------------------------------------|------------|--------|--|
| Course Name                                          | Code       | EC     |  |
| p1 Mathematical Foundations of Machine Learning      | XM_0189    | 6      |  |
| p1+2 Statistical Models                              | X_400418   | 6      |  |
| p2 Mathematical Optimization                         | XM_0051    | 6      |  |
| Choose at least 2 from:                              |            |        |  |
| p1+2 Applied Stochastic Modelling                    | X_400392   | 6      |  |
| p2 Dynamic Programming and Reinforcement Learning    | XM_0093    | 6      |  |
| p4 Large Language Models from scratch                | XM_0188    | 6      |  |
| p4+5 Optimization of Business Processes              | X_400422   | 6      |  |
| p4+5 Statistical Learning with High-Dimensional Data | XM_0206    | 6      |  |
| fall Asymptotic Statistics                           | Mastermath | 8      |  |
| fall Measure Theoretic Probability                   | Mastermath | 8      |  |
| spring Stochastic Processes                          | Mastermath | 8      |  |

| FURTHER RECOMMENDATIONS for electives                                                               |                                               | <=36 or 30 EC |    |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------|---------------|----|
| Course Name                                                                                         |                                               | Code          | EC |
| Match department's research profile,<br>will prepare you particularly well for your Master Project: |                                               |               |    |
| p2                                                                                                  | Performance of Networked Systems              | X_405105      | 6  |
| p1+2                                                                                                | Stochastic Processes for Finance              | X_400352      | 6  |
| fall                                                                                                | Continous Optimization                        | Mastermath    | 6  |
| fall                                                                                                | Forensic Probability and Statistics           | Mastermath    | 8  |
| fall                                                                                                | Time Series                                   | Mastermath    | 8  |
| fall                                                                                                | High-dimensional Probability                  | Mastermath    | 8  |
| spring                                                                                              | Machine Learning Theory                       | Mastermath    | 8  |
| spring                                                                                              | Scheduling Theory and Algorithms              | Mastermath    | 6  |
| spring                                                                                              | Queueing Theory                               | Mastermath    | 6  |
| Extra:                                                                                              |                                               |               |    |
| p4+5                                                                                                | Applied Analysis: Financial Mathematics (MSc) | XM_0140       | 6  |
| fall                                                                                                | Discrete Optimization                         | Mastermath    | 6  |
| spring                                                                                              | Causality                                     | Mastermath    | 8  |

Mathematics and Computation study path

| STRONG RECOMMENDATIONS for electives                               |            | >=22 EC |  |
|--------------------------------------------------------------------|------------|---------|--|
| Course Name                                                        | Code       | EC      |  |
| p1 Mathematical Foundations of Machine Learning                    | XM_0189    | 6       |  |
| Choose at least 2 from:                                            |            |         |  |
| fall Algebraic Topology I                                          | Mastermath | 8       |  |
| fall Dynamical Systems                                             | Mastermath | 8       |  |
| fall Partial Differential Equations                                | Mastermath | 8       |  |
| Mathematical Biology*                                              | Mastermath | 8       |  |
| spring Mathematical Neuroscience                                   | Mastermath | 8       |  |
| spring Elliptic Curves                                             | Mastermath | 8       |  |
| spring Formal Methods in Mathematics                               | Mastermath | 8       |  |
| Topological Data Analysis**                                        | Mastermath | 8       |  |
| *alternates with Mathematical Neuroscience, not offered in 2026-27 |            |         |  |
| **offered every other year, not in 2026-27                         |            |         |  |

| FURTHER RECOMMENDATIONS for electives, profile in Biomedical Mathematics                         |                                                 | <=44 or 38 EC |    |
|--------------------------------------------------------------------------------------------------|-------------------------------------------------|---------------|----|
| Course Name                                                                                      |                                                 | Code          | EC |
| Match department's research profile, will prepare you particularly well for your Master Project: |                                                 |               |    |
| p4+5                                                                                             | Statistical Learning with High-Dimensional Data | XM_0206       | 6  |
| fall                                                                                             | Measure Theoretic Probability                   | Mastermath    | 8  |
| fall                                                                                             | Numerical Methods for PDEs                      | Mastermath    | 8  |
| spring                                                                                           | Machine Learning Theory                         | Mastermath    | 8  |
| spring                                                                                           | Stochastic Processes                            | Mastermath    | 8  |
| Extra:                                                                                           |                                                 |               |    |
| fall                                                                                             | Functional Analysis                             | Mastermath    | 8  |
| spring                                                                                           | Inverse Problems in Imaging                     | Mastermath    | 6  |
| p1                                                                                               | From Molecule to Mind                           | AM_1275       | 6  |
| p1                                                                                               | Fundamentals of Bioinformatics                  | X_405052      | 6  |
| p1                                                                                               | Introduction to Systems Biology                 | X_428565      | 6  |
| p2                                                                                               | Algorithms in Sequence Analysis                 | X_405050      | 6  |
| p2                                                                                               | Mechanics and Thermodynamics in the Cell        | X_422589      | 6  |
| p4                                                                                               | Basic Models of Biological Networks             | X_418154      | 6  |
| p4                                                                                               | Brain Imaging (for AI)                          | P_MBRIMA_AI   | 6  |
| p4                                                                                               | Structural Bioinformatics                       | X_405019      | 6  |
| p5                                                                                               | Bioinformatics for Translational Medicine       | X_405092      | 6  |
| p5                                                                                               | Tracer Kinetic Modelling                        | XM_0014       | 6  |

| FURTHER RECOMMENDATIONS for electives,<br>profile in Algebra, Geometry and Number Theory            |            | <=44 or 38 EC |  |
|-----------------------------------------------------------------------------------------------------|------------|---------------|--|
| Course Name                                                                                         | Code       | EC            |  |
| Match department's research profile,<br>will prepare you particularly well for your Master Project: |            |               |  |
| fall Algebraic Number Theory                                                                        | Mastermath | 8             |  |
| fall Analytic Number Theory                                                                         | Mastermath | 8             |  |
| fall Modern Cryptography                                                                            | Mastermath | 8             |  |
| fall Algebraic Geometry I                                                                           | Mastermath | 8             |  |
| fall Advanced Algebraic Geometry                                                                    | Mastermath | 8             |  |
| fall Differential Geometry                                                                          | Mastermath | 8             |  |
| spring Algebraic Geometry II                                                                        | Mastermath | 8             |  |
| spring Algebraic Topology II                                                                        | Mastermath | 8             |  |
| spring Symplectic Geometry                                                                          | Mastermath | 8             |  |
| Extra:                                                                                              |            |               |  |
| fall Commutative Algebra                                                                            | Mastermath | 8             |  |
| spring Coding Theory                                                                                | Mastermath | 8             |  |
| spring Selected Areas in Cryptology                                                                 | Mastermath | 8             |  |

| FURTHER RECOMMENDATIONS for electives,<br>profile in Analysis and Dynamical Systems                 |                                               | <=44 or 38 EC |    |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------|---------------|----|
| Course Name                                                                                         |                                               | Code          | EC |
| Match department's research profile,<br>will prepare you particularly well for your Master Project: |                                               |               |    |
| fall                                                                                                | Numerical Methods for PDEs                    | Mastermath    | 8  |
| fall                                                                                                | Differential Geometry                         | Mastermath    | 8  |
| spring                                                                                              | Numerical Methods for SDEs                    | Mastermath    | 8  |
| spring                                                                                              | Non-Linear PDEs                               | Mastermath    | 8  |
| spring                                                                                              | Calculus of Variations                        | Mastermath    | 8  |
| spring                                                                                              | Interacting Particle Systems                  | Mastermath    | 8  |
| spring                                                                                              | Symplectic Geometry                           | Mastermath    | 8  |
| Extra:                                                                                              |                                               |               |    |
| p4+5                                                                                                | Applied Analysis: Financial Mathematics (MSc) | XM_0140       | 6  |
| fall                                                                                                | Functional Analysis                           | Mastermath    | 8  |
| spring                                                                                              | Inverse Problems in Imaging                   | Mastermath    | 6  |

| FULL LIST of ELECTIVES                                                                            |         |    |
|---------------------------------------------------------------------------------------------------|---------|----|
| Course Name                                                                                       | Code    | EC |
| - all local courses from above (code other than 'Mastermath')                                     |         |    |
| - an additional local course:                                                                     |         |    |
| p4 Entrepreneurship in Analytics and AI                                                           | XM_0090 | 6  |
| - all Mastermath courses                                                                          |         |    |
| - up to 2 level-300 courses from BSc Mathematics: advance permission from the Exam Board required |         |    |